Monitoring Plant Growth in Controlled Environments Using Image Processing

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Introduction

- Problem Statement : Develop an image processing system to monitor plant growth in controlled environments like greenhouses.

- Analyze plant health indicators, leaf count, and growth trends over time.

- Significance of Image Processing:

- Enables automated monitoring and control.

- Early detection of stress and diseases, improving crop health and minimizing losses.

Objectives

- Create an image acquisition model for plant monitoring.
- Develop preprocessing techniques to enhance image quality.
- Implement segmentation algorithms to isolate plant regions.
- Measure plant height and count leaves accurately.

Experimental Setup Controlled Environment:

- Small pots with a solid white background for contrast.

- Tripod-mounted camera for consistent angles.

- Image Capture Details:
  Resolution: 12.2 MP
  Frequency: 3 times/day (12 photos daily)

- Storage: Computer and external flash drive

- Lighting: LED light for night photography.

Preprocessing Pipeline Correct Non-Uniform Illumination:

- Techniques:
  Adaptive histogram equalization, homomorphic filtering.

- Remove Salt-and-Pepper Noise:
  Median filtering to clean pixel artifacts.

- Reduce Gaussian Noise:
  Bilateral filters or deep-learning denoisers for grainy noise.

- Edge Sharpening (Optional):
  Enhance details after distortion removal.

Image Segmentation Model

- Objective: Isolate plant regions for analyzing growth parameters like leaf count and structural changes.

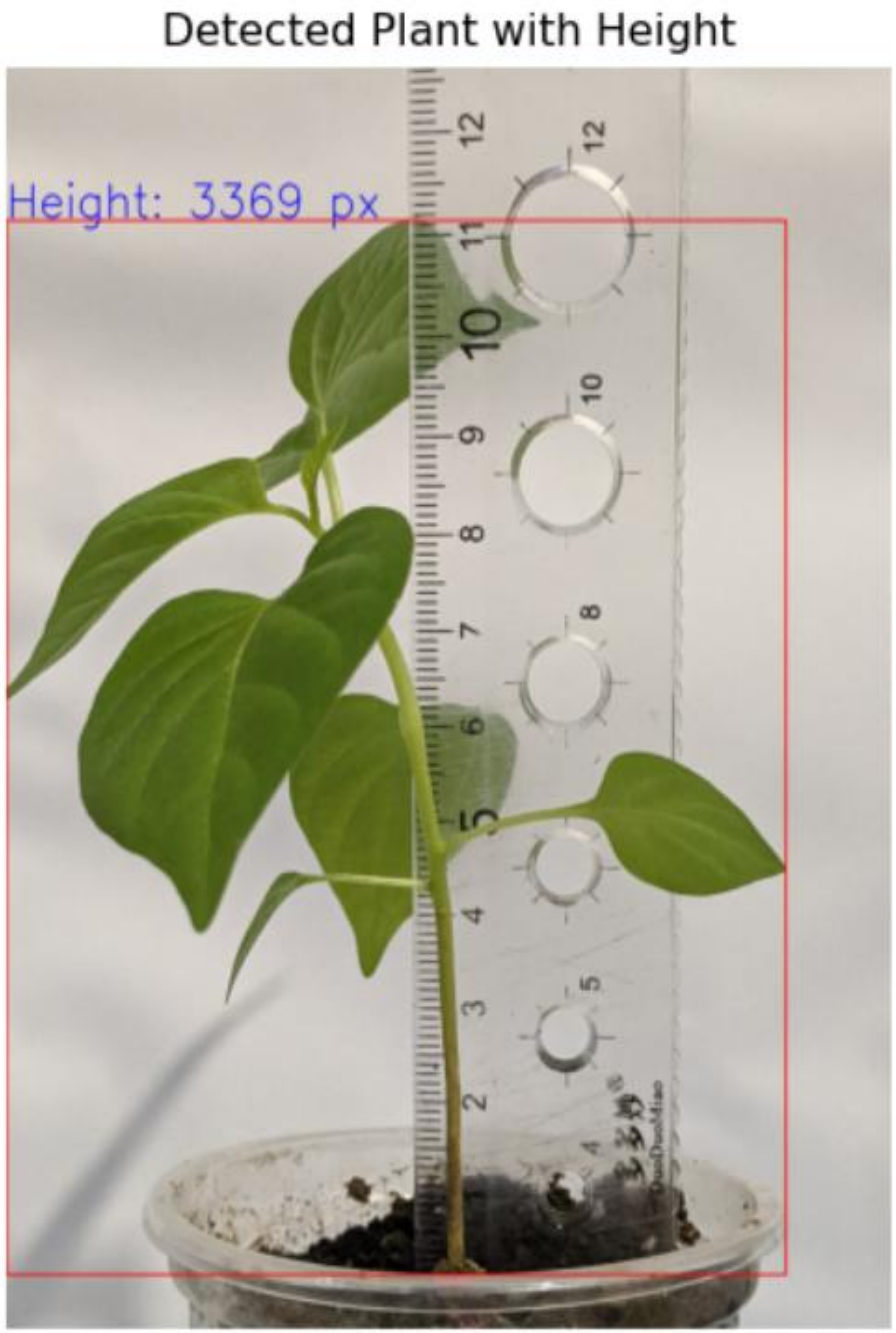
- Techniques Used: Color-based segmentation using HSV color space.

- Morphological operations to clean masks and isolate green areas.

Segmentation

- Workflow: Load Image → Convert to HSV Color Space.
- Define HSV range for green color → Create mask for green areas.
- Apply morphological operations → Clean noise and refine segmentation.
- Isolate plant regions from the background.

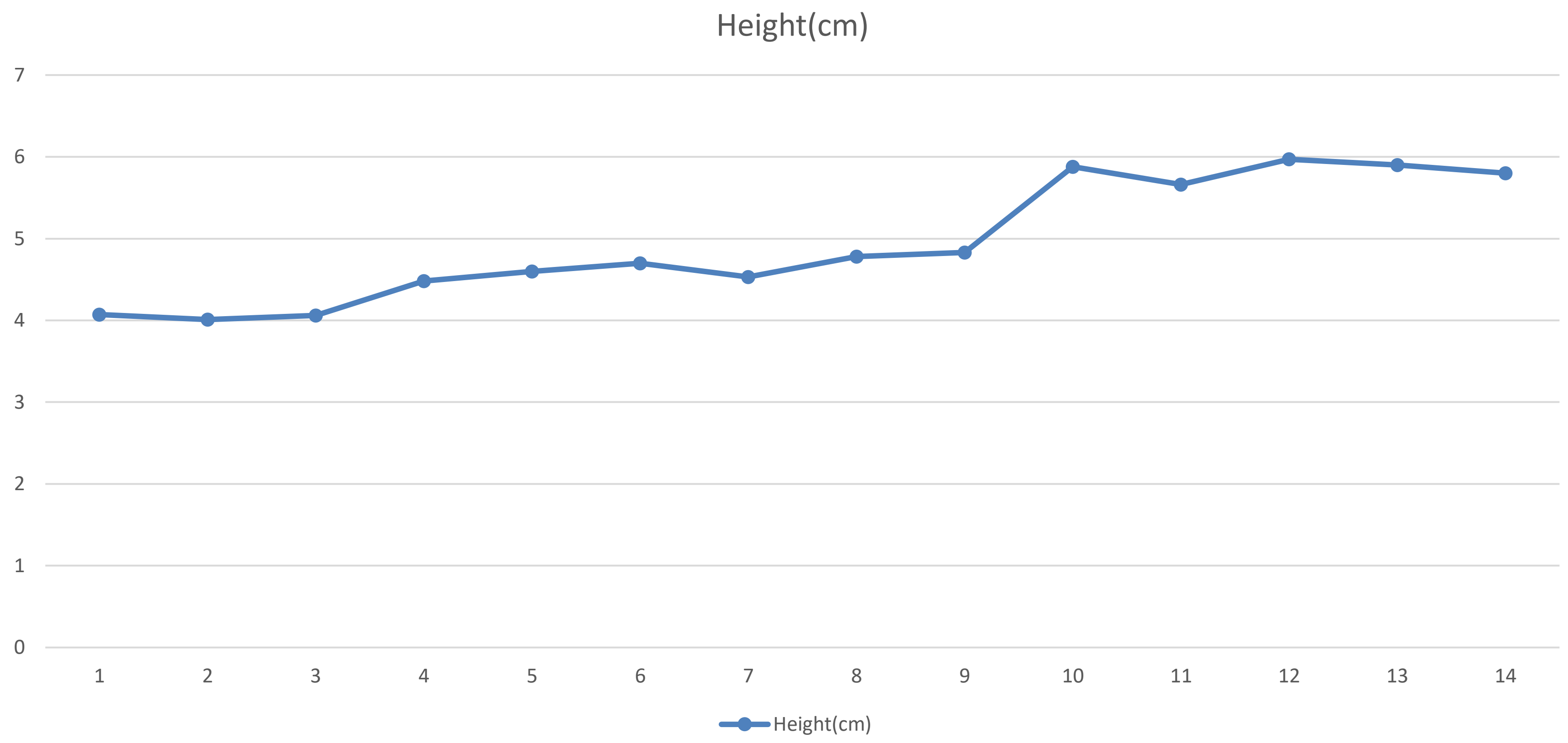
Plant Height Detection



- Measure plant height in pixels, then convert to centimeters using a pixel-to-centimeter ratio.
- Challenges:

  Variability due to leaf movement towards sunlight.

  Uniform camera-to-plant distance maintained for accuracy.



Leaf Counting Methodology

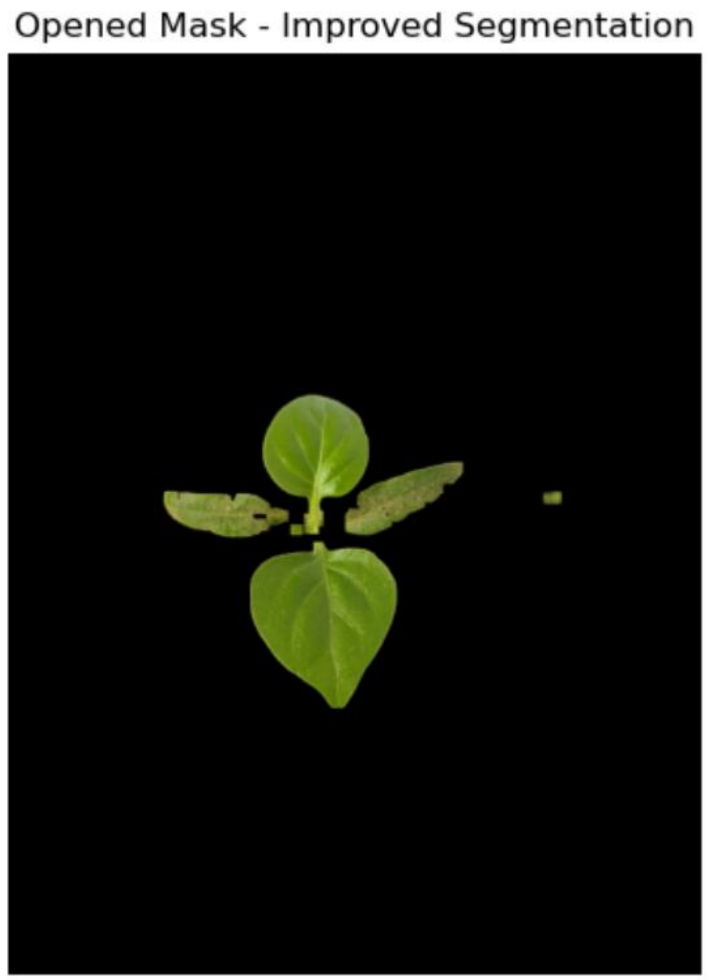
- Apply morphological opening to separate leaves from the trunk.

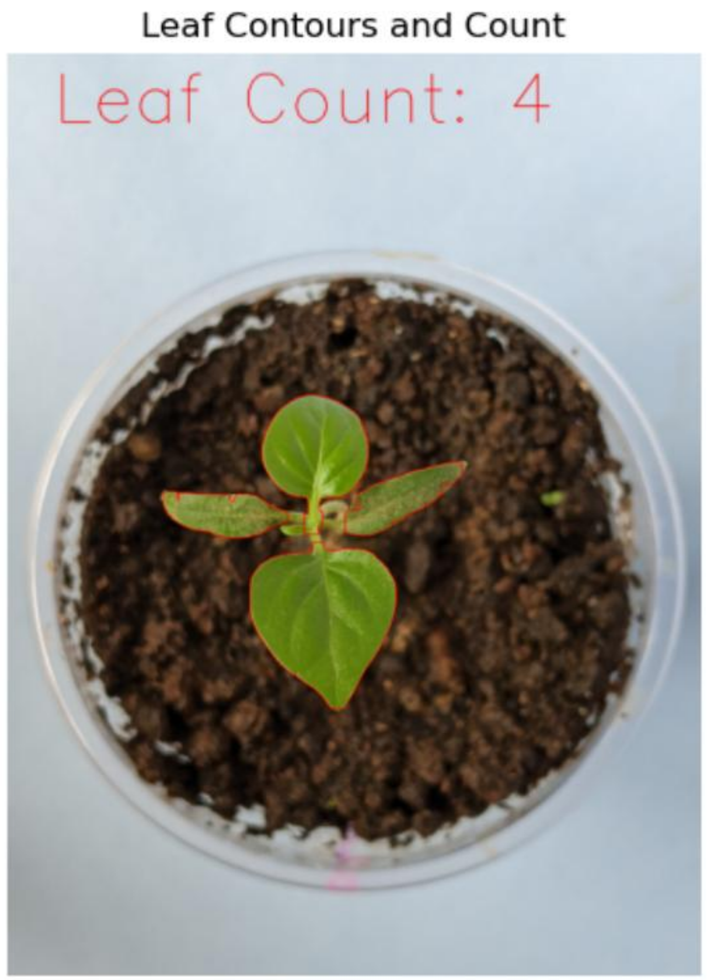
- Detect contours and filter based on area to remove noise.

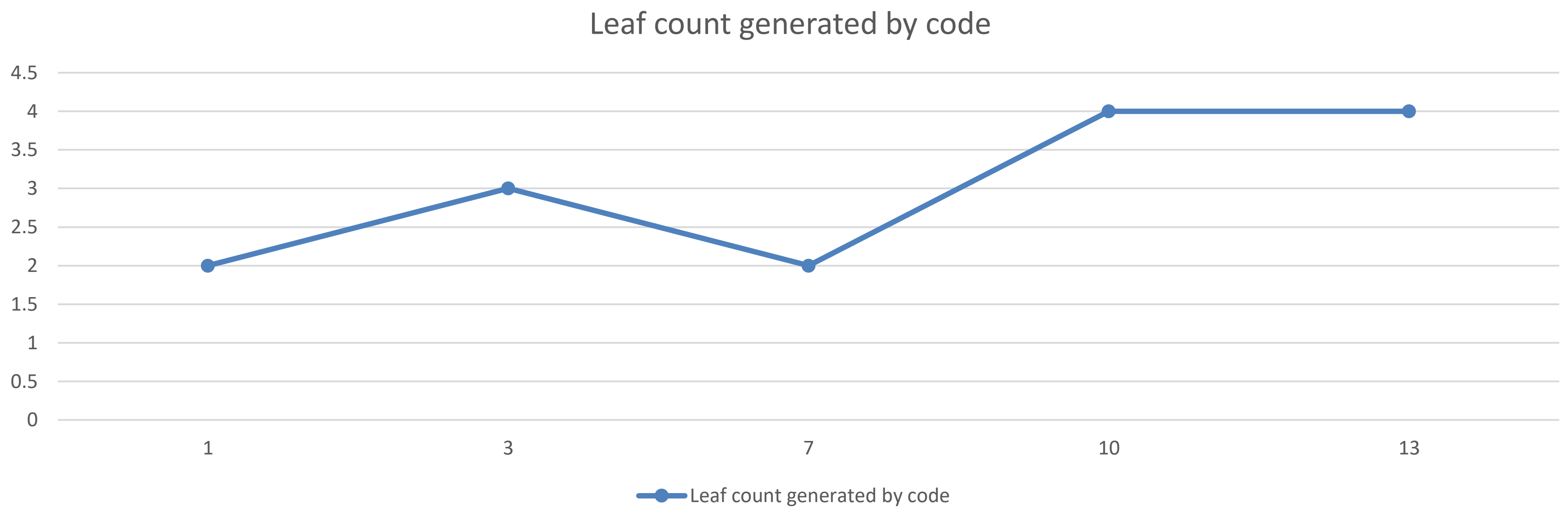
- Improve accuracy using:
  Gaussian blur for edge detection (Canny).

  Morphological operations (dilation, erosion) to refine edges.

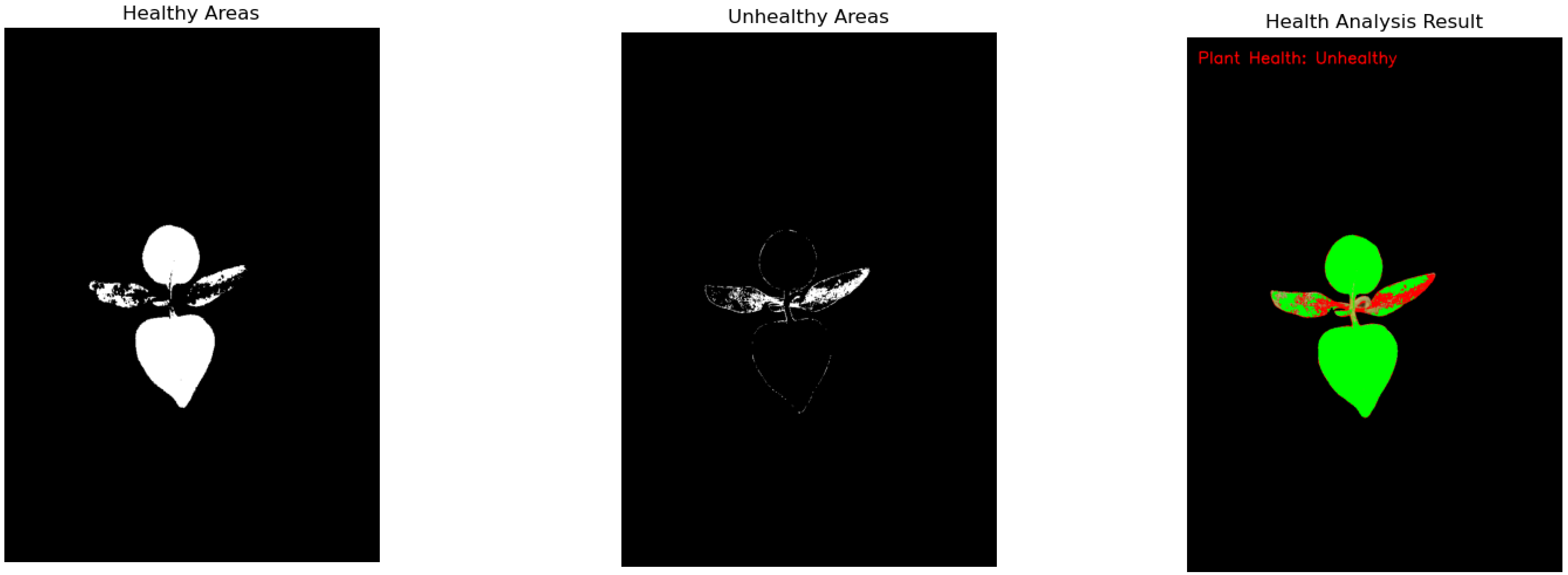
  Create a mask to isolate leaves and count them.







Plant Health Indication



- Healthy leaves are assumed to be predominantly green and are segmented using a specific HSV range.
- Unhealthy areas (yellow or brown) are segmented using another HSV range.
- Healthy regions are highlighted in green and unhealthy regions in red on the original image

Challenges and Limitations

- Artificial lighting required for night time photos.
- Variability in plant health due to inconsistent water/nutrition supply.
- Climate changes affecting plant growth.

Conclusion

- Image processing effectively monitors plant growth in controlled environments by:

  - Automating data collection and analysis.
  - Detecting early signs of stress or disease.
  - Limitations include environmental variability and overlapping leaves affecting measurements.

Acknowledgments & References

- https://www.youtube.com/watch?v=55eTVDto9K8&list=PLw7f_swC6ZDU5E4LN4bXXGpomHCIpGH4f
- https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html
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Q & A